Refining the Computational Analysis of Interactive Fiction: A Hybrid Framework using Bag-of-Paths Formalism and LLMs

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Introduction

Interactive Fiction and Gamebook

An **interactive fiction** is a form of narrative in which the reader's or player's choices influence the sequence of events, the accessible content, or the story's outcome.

A **gamebook** is a book-based form of interactive fiction, usually divided into numbered sections connected by choices, rules, and sometimes game mechanics such as combat, inventory, or chance.

112

Suddenly, the large rock you are hiding behind is rolled aside and you are faced by two snarling Giaks intent on your death. The cave mouth is a narrow entrance and you can only fight the Giaks one at a time.

Giak 1: COMBAT SKILL 13 ENDURANCE 10

Giak 2: COMBAT SKILL 12 ENDURANCE 10

If you win, you may explore the cave further by **turning to 33**.

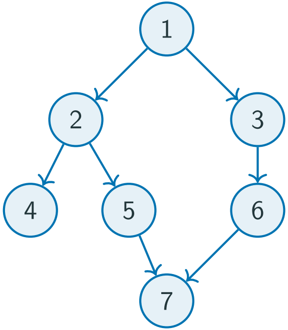
Or you may leave and descend the hill. **Turn to 248**.

Flight from the Dark by Joe Dever and Gary Chalk www.projectaon.org

The classical representation: a graph

Interactive fiction is traditionally modeled as a **directed graph**.

- **Nodes** represent *paragraphs*, i.e. numbered narrative sections.
- **Edges** represent *transitions*, i.e. the choices available to the player.



The **Main limitation** of this representation:

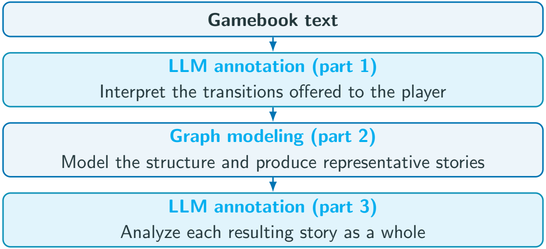
- The **structure** or the **topology** of the work can be studied.
- The text itself, therefore the **narrative meaning**, is no longer taken into account.

A hybrid solution: graphs and LLMs

Large Language Models (LLMs) make it possible to reintroduce the semantic content discarded by graph modeling.

Our approach: both methods

- The **graph** models the **structure** of the work
- The **LLM** interprets the **narrative content**



Research question

How can graph modeling and local LLM annotation be combined to study both the structural possibilities and the narrative experiences produced by a gamebook?

Methodological choice: local LLM

Methodological choice

- We are going to use exclusively **Qwen3.6-27B**, executed locally on university infrastructure.
- The local model is not treated as an autonomous literary critic. It is used as a **controlled annotation instrument** whose decisions remain open to inspection.

Auditable and independent

- Local execution on university infrastructure keeps source texts outside proprietary APIs.
- Prompts, inference settings, and outputs can be stored, inspected, and reproduced.

Appropriate to the task

- Annotation categories and admissible evidence are defined by an explicit codebook and JSON schema.
- A constrained annotation task does not require the generative capabilities of a frontier model.

Corpus and modeling scope

Project Aon provides authorized, HTML-structured editions of Joe Dever's *Lone Wolf* series. This proof of concept studies only **Book 1: Flight from the Dark**.

The objective is a tractable **Markovian model**: once the profile is fixed, the next transition depends on the current paragraph rather than the complete play history.

Level	Information	Mechanics	Decision
L0	Topology	Paragraphs, links, forced transitions, Death and Win outcomes	**Included**
L1	Player agency	Choices annotated along the risk, morality, and action axes	**Included**
L2	Uncertainty	Random events, combat, evasion, Kai Disciplines, and conditions	**Simplified**
L3	Persistent state	Endurance, healing, inventory, gold, equipment, and meals	**Not simulated**

L2 uses explicit, global probabilities. Their values are reported when the profile-dependent graph is introduced.

Paragraph parsing and transition annotation

Objective: from paragraphs to structured data

Objective

Transform each gamebook paragraph into an **structured representation**, separating **narrative units** from the **transitions** offered to the player.

Project Aon page

112

Suddenly, the large rock you are hiding behind is rolled aside and you are faced by two snarling Giaks intent on your death. The cave mouth is a narrow entrance and you can only fight the Giaks one at a time.

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Paragraph 112

$\Longrightarrow$

Parsing

+

LLM

LW01_nodes.csv

node_id	text_content	absorbing_status	enemies
112	Suddenly, the large rock...	potential_death	Giak 1; Giak 2

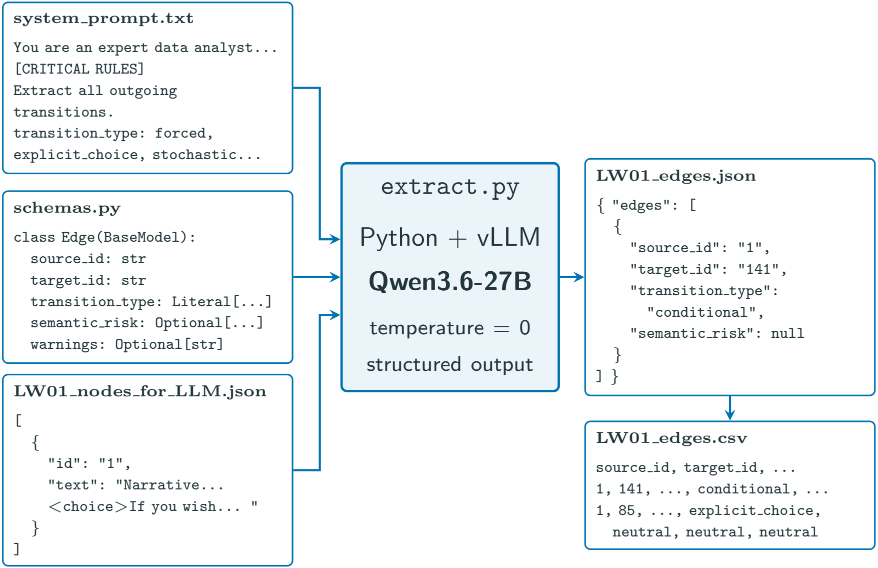
LW01_edges.csv

source_id	target_id	edge_text	transition_type	realisation_value
112	33	If you win, explore the cave...	explicit_choice	If you win
112	248	Leave and descend the hill...	explicit_choice	If you win

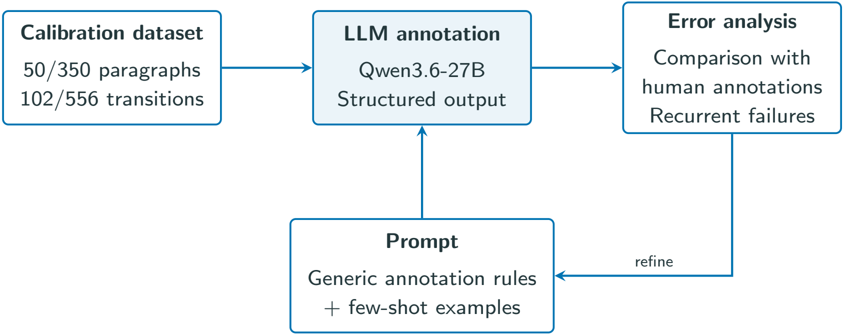
source_id	target_id	semantic_risk	semantic_morality	semantic_action	warnings
112	33	reckless	neutral	tactical	–
112	248	cautious	selfish	physical	–

Each explicit choice is annotated on three independent axes: **risk** (cautious–reckless), **morality** (selfish–noble), and **action** (physical–tactical); each axis also allows a neutral value.

From prompt to structured annotation



Iterative prompt refinement



- Errors lead to revisions of **general annotation rules**, not paragraph-specific corrections.
- This iterative process is a **calibration**, not an independent evaluation.
- No held-out validation set was used because of limited time; generalization to another book is therefore not measured.

Annotation results

Calibration: On 50 paragraphs and 102 transitions, the final prompt produced no missing or invented transition and no transition-type disagreement. Only 3 (of 177 annotations) semantic disagreements: two for risk and one for action.

Paragraph statistics

Total	350
Terminal	16 deaths / 1 victory
One outgoing transition	157
Two outgoing transitions	134
Three or more outgoing transitions	42
Paragraphs with enemies	29

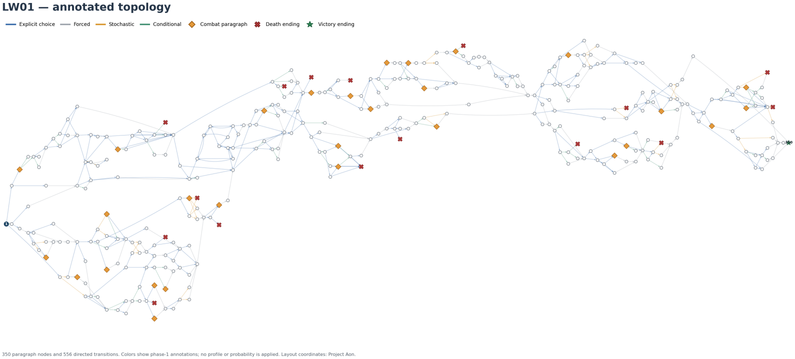
Transition statistics

Transition type	N	%
Explicit choice	291	52.3
Forced	157	28.2
Conditional	66	11.9
Stochastic	42	7.6
Total	**556**	**100.0**

Axis	First pole	Neutral	Opposite pole
Risk	cautious: 95 (32.6%)	84 (28.9%)	reckless: 112 (38.5%)
Morality	selfish: 18 (6.2%)	262 (90.0%)	noble: 11 (3.8%)
Action	physical: 101 (34.7%)	126 (43.3%)	tactical: 64 (22.0%)

Warnings	53 (9.5%)

Result of paragraph annotation : unweighted graph



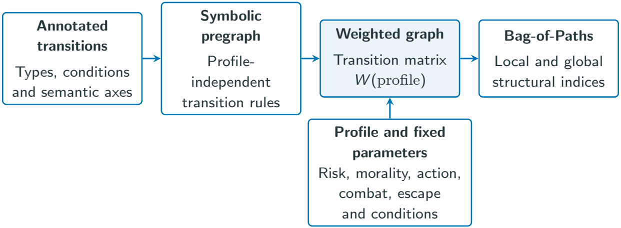
The topology and transition types are now available, but **transition probabilities depend on player profiles** (next section).

Graph construction and Bag-of-Paths analysis

Objective: from annotations to structural measures

Objectives

- Transform annotations into **weighted directed graphs (i.e. a Markov chain matrix)** which depend on **player profiles**.
- Compute **various indices with the Bag-of-Paths formalism**.



Questions:

- How can game mechanics be converted into transition probabilities?
- How does the resulting graph vary across player profiles?
- Which parts are most important across all possible trajectories?

One Markov chain per player profile

The matrix $\mathbf{W}^{(p)}$ **describes the behavior of one fixed player profile**. Once the profile and the experimental parameters are fixed, the next transition depends only on the current paragraph (Markov chain).

Player profile

$$p = (\text{risk}, \text{morality}, \text{action})$$

- Risk: cautious, neutral, reckless
- Morality: selfish, neutral, noble
- Action: physical, neutral, tactical

$3^3 = 27$ **player profiles**

Fixed parameters (all profiles)

- Combat victory: $P(\mathrm{win}) = 0.833$
- Combat evasion: $P(\mathrm{escape}) = 0.5$
- Required Kai Discipline available: $P(\mathrm{Kai}) = 0.5$
- Item, gold, or Endurance condition satisfied: $P(\mathrm{has\_condition}) = 0.5$

From profile to probability

Each choice annotation is compared with the profile:

Relation	Affinity
Matching non-neutral values	2
Neutral profile or annotation	1
Opposed non-neutral values	0.5

Affinities are multiplied and normalized among the available outgoing choices.

For a reckless profile facing one reckless and one cautious choice:

$$P(\mathrm{reckless}) = \frac{2}{2+0.5} = 0.8$$

$$P(\mathrm{cautious}) = 0.2.$$

Bag-of-Paths at the Random-Walk limit

For each player profile, $\mathbf{W}^{(p)}$ represents the corresponding Markov chain. For the BoP calculation, absorbing states have no outgoing transitions: $W^{(p)}$ is therefore **substochastic**, and we can compute:

$$\boxed{\mathbf{Z}^{(p)} = \left(I - \mathbf{W}^{(p)}\right)^{-1} = I + \mathbf{W}^{(p)} + \left(\mathbf{W}^{(p)}\right)^{2} + \cdots}$$

$z_{si}^{(p)}$ is the **expected number of visits to paragraph i when starting from paragraph s**. Many indices can be derived from the same fundamental matrix:

Local indices

Computed for individual paragraphs or transitions:

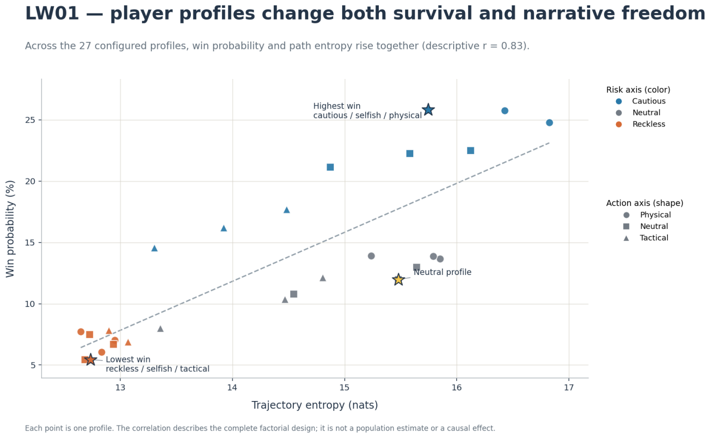
- Probability of visiting a paragraph
- Contribution to mortality
- Sensitivity to the player profile

Global indices

Computed for the complete book and one profile:

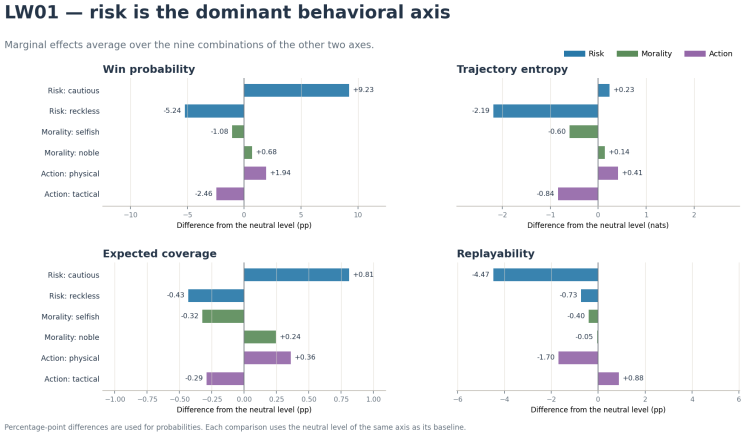
- Probabilities of victory and death
- Expected trajectory length and entropy
- Divergence between player profiles

Player profiles change survival and narrative freedom



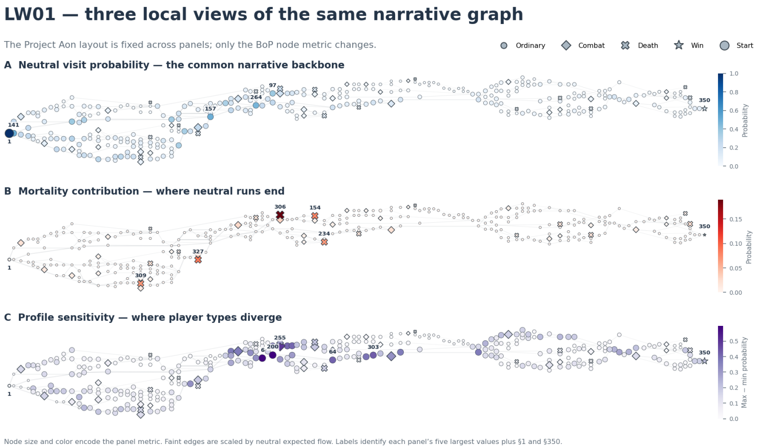
Across the **27 configured profiles**, win probability ranges from **5.4% to 25.8%**. Survival and trajectory entropy rise together ($r = 0.83$): profiles that survive longer also access a wider range of paths.

Risk is the dominant behavioral axis



Moving from neutral to cautious choices increases win probability by **9.23 percentage points**; moving to reckless choices reduces it by **5.24 points**. Morality and action have smaller, but non-zero, effects.

Different indices reveal different key paragraphs



The common narrative backbone, the main mortality bottlenecks, and the regions where player profiles diverge **do not coincide**. Local importance therefore depends on the structural question being asked.

Trajectory annotation

Objective: from probabilistic paths to complete stories

Objective

See if player profiles become perceptible in **complete stories**.

1. Path selection
7 profiles × Win/Death
2,000 paths per cell
14 empirical medoids

→

2. Blind corpus
Complete story text
Choices encountered
No player profile
14 texts

→

3. LLM annotation
Local Qwen3.6-27B
Individual annotations
Pairwise annotations
Constrained JSON

→

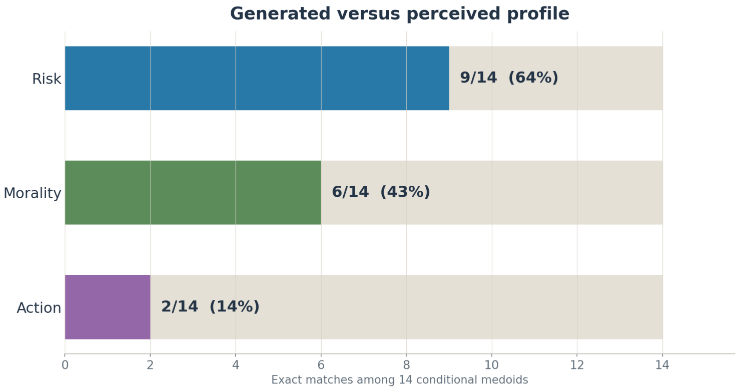
4. Analysis
Profile manifestation
Causal continuity
Profile coherence
Controlled contrasts

A **medoid** is an observed trajectory that minimizes its average distance to all other sampled trajectories, making it the most central representative path.

The model infers whether the intended profile becomes perceptible from the **story and choices only**. The hidden metadata and BoP measures are reintroduced only for the final comparison.

Calibration: the phase-1 refinement loop was reused on 4/14 stories and 3/6 pairs. The third generic prompt was frozen after 7/7 valid outputs, with 32/44 human–model concordances.

Individual annotation results



Causal continuity	Profile coherence	Action recovery by level
14/14	9/14	Neutral 0/10; Physical 0/2; Tactical 2/2

Pairwise annotation results

Profile contrast	Outcome	Path difference (LCS)	A on Risk	A on Morality	A on Action	Narrative distinctness
Cautious (A) vs reckless (B)	Win	0.61	**cautious**	selfish	tactical	High
Cautious (A) vs reckless (B)	Death	0.79	**cautious**	selfish	tactical	High
Selfish (A) vs noble (B)	Win	0.41	cautious	**selfish**	tactical	Medium
Selfish (A) vs noble (B)	Death	0.21	cautious	**selfish**	tactical	Medium-high
Physical (A) vs tactical (B)	Win	0.38	even	noble	**even**	Medium-high
Physical (A) vs tactical (B)	Death	0.30	reckless	even	**physical**	Medium

Reading: bold = controlled axis; red = A/B–B/A disagreement averaged across orders, "main tendency" output displayed.

Intended profile contrast recovered in **5 of 6** comparisons.

Conclusion

Conclusion

What we learned

- Profiles reshape the probabilistic structure.
- Risk dominates the other axes.
- 5/6 relative story contrasts are recovered; absolute profiles remain unevenly visible.

Current limits

- One gamebook only.
- Simplified mechanics and state for the Markov model.
- Limited human calibration; no held-out validation.

Next

- Test the workflow comparatively on other books.
- Add richer game-state mechanics and independent human validation.

This is a **proof of concept for an auditable hybrid workflow**: BoP measures narrative structure, while the local LLM remains a controlled annotator rather than a literary critic.

Thank you for your attention!

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